

Github repo:

github.com/paulbradshaw/mass_foi_projects

Workshops in the **[walkthroughs folder](#)**

No.	Financial Year	Crime Type	Outcome Description
242	2019/2020	VIOLENCE AGAINST THE PERSON	Type 16 - Named Suspect - Victim Unsupportive
243	2019/2020	THEFT	Type 18 - No Suspect Identified
244	2019/2020	VIOLENCE AGAINST THE PERSON	Type 16 - Named Suspect - Victim Unsupportive
245	2019/2020	BURGLARY	
246	2019/2020	THEFT	
247	2019/2020	THEFT	

Crime	2019	2020	2021	2022
-------	------	------	------	------

Crime Sub Group	Total
-----------------	-------

FOI+AI

Mass FOI projects in an AI era

Name	
 Bedfordshire.docx	Microsoft Word document (.docx)
 Cleveland (downloa...open in outlook).msg	Document
 essex.xlsx	Microsoft Excel Workbook (.xlsx)
 PSNI_response.csv	Comma Separated Spreadsheet (.csv)
 Sussex.pdf	PDF Document

Residential Burglary of a Home	1
--------------------------------	---

The scenario:

- Claudia sent an FOI to every police force
- She got a whole mix of responses back
- We needed to be able to turn that into a single dataset that could be questioned for a story

In 1hr you will be able to

- Manage **mass FOI projects** more efficiently
- Use vibe coding to **clean and standardise** FOI responses
- **Verify** the results

Comments welcome

This is work in progress — your questions, comments and suggestions will help identify blind spots, risks and improvements.

Vibe coding alone? Fail!

ChatGPT Codex: “I have a folder of FOI responses with data about crimes in hospitals. Help me get them into a consistent shape”

Result: **half of incidents and one third of forces missing, with incidents missing for 13 of the remaining 20 forces.**

What we need is a proper methodology

Audit & prioritise

Audit responses to identify the level of detail in each response and identify edge cases. [Include a caveats column.](#) Augment manual audit with **NotebookLM** audit. **Identify a priority order** for data, e.g. totals by outcome, hospital, category or year where these are provided separately

Data structure

Design a data structure that can accommodate all responses
Structure should follow **'tidy' data principles**, i.e. one row per combination of features (force, category, hospital, outcome, year)
Structure should include source details, e.g. filename, sheet name, name of person entering data

Extract & reshape

PDFs: use **Tabula** or **vibe coding** (design a prompt template to generate code to attempt to extract data). Multi-sheet XLS files: use **Open Refine** to import and combine sheets
Design a **prompt template** for generating code to reshape CSV responses

Check & verify

Manual checks (e.g. compare entries, check page-ending rows)
Analysis-based checks (e.g. pivots, totals)
AI-based checks using a prompt template (e.g. compare files)

Combine

Use **OpenRefine** or: **Design a prompt template** for generating code to combine the resulting CSV files.

Problem 1:
The responses are
inconsistent

No.	Financial Year	Crime Type	Outcome Description
242	2019/2020	VIOLENCE AGAINST THE PERSON	Type 16 - Named Suspect - Victim Unsupportive
243	2019/2020	THEFT	Type 18 - No Suspect Identified
244	2019/2020	VIOLENCE AGAINST THE PERSON	Type 16 - Named Suspect - Victim Unsupportive

245	2019/2020	BURGLARY
246	2019/2020	THEFT
247	2019/2020	THEFT

	Crime	2019	2020	2021	2022
248	All other theft offences	58	59	117	293
249	Bicycle theft	3	8	9	22
251	Business and community burglary	4	11	18	24
2020-03	Criminal damage and arson offences	36	46	121	202

Crime Sub Group	Total
Arson	19
Bicycle Theft	92
Burglary Business & Community	63
Burglary Residential	2
Criminal Damage	458
Homicide	1
Miscellaneous Crimes Against Society	28
	190
	463
	88
	47
	775
	45
Residential Burglary of a Home	1

Name	Kind
 Bedfordshire.docx	Microsoft Word document (.docx)
 Cleveland (downloa...open in outlook).msg	Document
 essex.xlsx	Microsoft Excel Workbook (.xlsx)
 PSNI_response.csv	Comma Separated Spreadsheet (.csv)
 Sussex.pdf	PDF Document

Break it down: each inconsistency is a separate task. What needs doing first?

1. **Level of detail:** prioritise (choose a level to focus on)
2. **Filetype:** extract data into one format (CSV)
 - a. Combine sheets from XLSX
 - b. Extract table from DOC
 - c. Extract table from PDF
3. **Data shape:** reshape data (wide to long, all fields)
4. **Categories/names:** clean data (standardise)
5. **Timescale:** report accurately with caveats (edit)

Problem 2:
You need to choose
a focus

Audit the FOIs

- **Browse the FOIs**, or a representative sample
- **Identify the range** of information covered
 - Are there any questions which some bodies refused to answer?
 - Or levels of detail they couldn't provide?
- **Identify any differences** (e.g. financial year vs calendar year)
- **Decide what takes priority** where multiple tables are given (e.g. offence category, not hospital or outcome)

Force	Format	Hospital	Category	Outcome	Year	Year type	Also available in response	CAVEATS
Bedfordshire	DOC	N	Y	N	Y	Financial	Outcomes by year	
Cambridgeshire	DOC	N	Y	N	Y	Financial	Outcomes by year	
City of London	XLS	Y	Y	Y	Y	Financial		3 crimes have no year. 1 crime is dated January 1993
Cleveland	MSG	Y	Y	Y	Y	Financial	Offences by year	
Cumbria	PDF	Y	N	Y	N	5 years	Categories by year	
Durham	PDF	Y	N	N	Y	Financial	Categories by year; outcomes by year	
Dyfed-Powys	DOC	Y	Y	Y	Y	Calendar		Two different systems used for 2023; different system for 2024. Hospitals and wards inconsistently named - extra cleaning needed?
Essex	MSG	N	N	Y	N	Financial	Offences by year	
Hertfordshire	PDF	N	Y	Y	Y	Financial		
Humberside	PDF	N	Y	N	Y	Calendar	Outcomes by year	
Kent	XLS	Y	Y	Y	Y	Financial		
Leicestershire	XLS	N	Y	N	Y	Financial	Outcomes by year; hospital by year	Only uses codes

Tip: Prompt template for NotebookLM

OBJECTIVE: You are a journalist looking to combine data from multiple FOI responses into a single table that can be analysed to identify trends over time and compare categories or bodies.

TASK: Audit these responses and produce a table identifying the range of information covered for each force. Each column should identify Y/N if they provided that information.

Include a column indicating what type of year was used (e.g. financial vs calendar) and a column indicating whether data is provided for each incident, or per year, month, quarter, another period.

If the response includes any warnings or caveats, quote these and the page number in a caveats column.

**Copilot has a limit of 3 documents so cannot do this*

Try it now.

Force Name	Total Crimes Provided	By Year	Crime Nature	Crime Outcome	Hospital Name	Year Type	Period Frequency	Notes
Cumbria	Y	Y	Y	Y	Y	Financial	Per Year	"this force's response to your questions should not be used for comparison purposes with any other response you may receive." (p. 9)
Durham	Y	Y	Y	Y	Y	Financial (implied)	Per Year	"there were 0 crimes recorded at some of the locations of interest for the timeframe of the request therefore do not feature in the tables." (p. 1)
Greater Manchester	Y	Y	Y	Y	N	Financial	Per Year	"We cannot confirm if the crime occurred in or on hospital grounds or the hospital name without manually reviewing each crime to confirm the information." (p. 18)
Hertfordshire	Y	Y	Y	Y	Y	Financial	Per Incident	N/A in sources provided.
Humberside	Y	Y	Y	Y	N	Financial (implied)	Per Year	"the systems used for recording these figures are not generic... response to your questions should not be used for the comparison purposes with any other response you may receive." (p. 51)
Norfolk & Suffolk	Y	Y	Y	Y	Y	Calendar	Per Year	N/A in sources provided.
North Wales	Y	Y	Y	Y	Y	Fiscal	Per Incident	"The name of the hospital is not always recorded." (p. 58)

Problem 3:
You need to
extract the data

Things to try **before** using AI:

- **Try copy and paste** with DOC and emails (and XLSX)
- **Try Open Refine** to combine sheets from XLSX — or search DuckDuckGo* to find other solutions like this
- **Try Tabula** first for PDF extraction: tabula.technology
- **Splitting/combining** PDFs if too large/many: Adobe Acrobat
- **Security assessment:** are the docs sensitive?

All the above are **quicker and less energy intensive** than AI, and **reduce deskilling and other risks**

**You can turn off AI in DuckDuckGo but not in Google*

Prompt template: XLSX extraction

This XLS file contains a sheet for each financial year. Each sheet has three tables.

Extract the second table, in columns D-E, from each sheet, and create a CSV containing the combined tables with the following columns:

Offence Code | Total | Year from | Year to

Ignore the first two sheets which do not relate to financial years

Prompt template: XLSX extraction

This XLS file contains a sheet for each financial year. Each sheet has three tables.

Provide context:
what is in the sheets

Extract the second table, in columns D-E, each sheet, and create a CSV containing the combined tables with the following columns:

Name specific
columns/rows

Offence Code | Total | Year from | Year to

Specify output +
specify structure

Ignore the first two sheets which do not relate to financial years

Negative
prompting: tell it
what NOT to do

Prompt template: PDF extraction

Attached is a PDF. Extract ONLY the table on pages 2 and 3 which shows crime totals by category and year. *If any line breaks or hyphenations cause split labels, reconstruct using nearest-neighbor/line-merge logic and re-validate.*

Export as a CSV.

Check you have grabbed rows at the ends and beginnings of pages.

Check that figures in the extracted CSV, when totalled match totals in the PDF

If any uncertainty remains (e.g., an OCR-confused label), include a "flags" list with the exact text span and your best-guess normalization.

Prompt template: PDF extraction

Attached is a PDF. Extract ONLY the table on pages 2 and 3 which shows crime totals by category and year. *If any line breaks or hyphenations cause split labels, reconstruct using nearest-neighbor/line-merge logic and re-validate.*

Export as a CSV.

Check you have grabbed rows at the ends and beginnings of pages.

Check that figures in the extracted CSV, when totalled match totals in the PDF

If any uncertainty remains (e.g., an OCR-confused label), include a "flags" list with the exact text span and your best-guess normalization.

Name specific pages + tables

Build checks in for line breaks + page breaks.

Include validation checks

Allow room for uncertainty + checks

Using AI: **extract** — or **code**?

You can ask AI to **extract** the table from the PDF but...

You can also ask for **code** you can run (“in Google Colab”) that will generate the same result.

This has the advantage of **explainability**, aids **verification**, and reduces **deskilling**.

Prompt template for Copilot/ChatGPT/etc.

This is going to be a long prompt, so break it down:

Objective

Task

Table structure

Extra details

Prompt pt1: Overall objective/context

Objective:

To import data from an FOI response containing data on crimes in hospitals and clean it so that we have a single dataset in a common structure.

The **order of priority** is as follows: ideally data which shows every crime, its category, outcome and location.

But if this is not in the response, we want any table showing total crimes by category for each year - or for all years.

If that isn't provided, total crimes by hospital and year - or for all years.

If that isn't provided, total crimes by outcome and year - or for all years.

Prompt pt2: Break down the task

Task

Look at the attached PDF and suggest Python code that will work in Colab notebook. It should:

1. Import the PDF
2. Identify the pages with data to extract. Follow this order of priority: 1) Detailed data on each individual crime, including information on category, outcome and/or location; 2) Aggregate data on crimes by category; 3) crimes by hospital; 4) crimes by outcome.
3. Extract that data into a series of dataframes. Store the force, filename and page number(s) in extra columns. Note that tables which continue into subsequent pages will start with a row of data, rather than the headings.
4. Convert it into a single dataframe with headings from the first table (because where tables continue into subsequent pages they won't have the headings on them)
5. Clean the dataframe to use the field headings from that table.
6. Export and download the CSV

Prompt pt3: Describe the PDF

Table structure [DELETE/ADAPT AS APPLICABLE]

Here is the structure of the tables in the PDF.

The table appears on page XXX under question XXX and continues onto the first part of page XXX

In the PDF there are grid lines between the rows and columns in the table.

The first column is XXXXX, then there are XX columns for each year, and a grand total column

[PASTE EXTRACT]

Prompt pt4: Extra details (optional)

Nottinghamshire: in this response the police force has provided a separate table for each hospital. Each table contains a first column titled 'year' with the categories of crime in the rows underneath, then five columns with figures for each of five financial years. The name of the hospital appears just above the table.

Grab each table including the name of the hospital above it. Put this in a column called 'hospital'

Ignore the outcomes tables that come immediately after each of those.

Paste it into a Colab notebook

1. Go to colab.research.google.com/ and create a **new notebook**
2. In ChatGPT/etc. **use the ‘copy’ button** in the upper right corner of the code
3. **Paste it into the code block** in Colab and click the ‘run’ button
4. **Look for an ‘Upload’ button** appearing underneath the code block, where you can upload the PDF

If the code generates an error, paste it back into AI or use Colab’s built-in AI. If the data isn’t quite right, delete what you don’t want, or describe your problem.

Try the
walkthroughs

**Move onto the
verification, reshaping
and cleaning
walkthroughs
if you want**

Problem 4:
You need to check
it's accurate

Cross-check extracted data against original

Combine three methods:

1. **“Eyeball” tests:** manual checks for known risks, vulnerabilities and issues
2. **Analysis tests:** calculations such as totals and value counts to cross-check
3. **AI tests:** document comparison and repeat of above tests

Check	Completed by	Notes/action taken
Blanks/merged cells in original: check values haven't shifted		
Multi-line cells: check for no truncation		
First/last rows on PDF pages: check values extracted		
Headers & labels: check extracted and match data type		
Totals: check they are extracted/omitted as needed		
Non-numeric data: ("–", "<5"): extracted accurately		
Spot-checks: for individual values at random		
Values count: actual numbers vs expected (e.g. 5 cols x 8 rows)		
Non-numeric data: ("–", "<5"): impacts on totals/counts		
Row totals*: calculate from values & match to PDF totals		
Column totals*: calculate & match to PDF		
Pivot table*: construct & match to PDF		
Sequence/ID continuity: check for gaps using formulae		
AI check: value matching		Template here
AI check: row matching		
AI check: total checks		

Walkthrough.

Prompt template pt1: checking (new chat*)

**Make sure memory is turned off between chats*

CONTEXT

Attached is a CSV and the source PDF.

Focus only on the table(s) specified below and perform all of the following checks.

Report findings by section.

Target table: [DESCRIBE LOCATION/LABELS/STRUCTURE]

Ignore any tables that [DESCRIBE OTHERS]

If you cannot read all pages of the PDF (e.g., due to encryption or rendering limits), explicitly state which pages were validated and confine all findings to those pages. Recommend next steps for validating the remaining pages (e.g., obtaining a non-encrypted copy or exporting page by page).

Conclude with a one-line verdict: Accurate – no discrepancies found across all checks, or Discrepancies found – followed by a brief summary of the most significant issues.

Prompt pt2: value matching

CHECKS

1. Value matching

Compare every individual cell value between the PDF table and the CSV.

Report any numerical discrepancies as: *PDF value → CSV value* with row label and column/year.

Identify every instance where the PDF shows non-numerical values such as `–` or `<5` alongside numerical values. For each, state how it appears in the CSV, and assess whether the representation affects totals (e.g., `<5` stored as text vs a numeric value).

Note cosmetic differences separately (header typos, spacing, em-dash vs zero, etc)

Prompt pt3: row matching

2. Row matching

Flag any rows or categories present in the PDF but missing from the CSV, or extra rows in the CSV not in the PDF.

Check for systematic gaps in the CSV relative to the PDF, such as one missing row per page or missing first/last rows at page breaks.

If the PDF contains sequential IDs or numbering, list any missing IDs, infer any pattern (e.g., every 59th row absent), and provide examples.

Prompt pt4: total checks

3. Total checks

Compute the totals for the CSV and check those exactly match totals in the PDF (calculate those if not printed)

Do this for both row totals, column totals, and any other totals

Report any mismatches with the relevant row label and year column(s) .

Prompt pt5: label checks

4. Label checks

Compare all category, group, or label strings between the PDF and CSV.

Identify mismatches caused by punctuation, whitespace, or character encoding differences (e.g., curly apostrophes vs straight, en-dash vs hyphen).

For each discrepancy, map the PDF label to its CSV variant, estimate how many rows are affected, and confirm whether the underlying numeric values are otherwise aligned.

Prompt pt6: count checks

5. Count checks

Check that the CSV has the expected number of values based on the number of rows and columns in the PDF, excluding grand totals (e.g. 5 columns $\times$ 8 rows = 40 values)

Count how many values are in each row in the CSV, and how many are in each row in the PDF. Identify any mismatches.

What AI checks picked up

1. **Manually entered data** that contained errors (e.g. autocomplete wrongly completed)
2. **Artefacts of cleaning** (e.g. dashes removed in categories, or categories split on the dash)

Problem 5:
You need to get
the responses into
one shape

Decide on a common data structure

You will need: a structure that can **accommodate the variety of responses**, and that **allows you to ask different questions** (e.g. pivot tables)

1. **Have ‘year from’/‘year to’** (DON'T try to have five columns for five years)
2. Have a column that measures the **number** of events (e.g. crimes)
3. **Have a category column** that classifies the type of event (e.g. ‘theft’)
 - a. You might have sub-category column as well
4. **Allow columns to have ‘NOT SPECIFIED’** where bodies didn't provide that level of detail (or combined detail)
5. **Have columns for filename and authority** for traceability/combination

Bad structure

**Year should be stored
as data, not a heading**

Crime Volume with Location containing 'Hospital'	2019/2020	2020/2021	2021/2022	2022/2023	2023/2024
104 - Assault Without Injury on a Constable	12	29	29	35	51
105A - Assault Without Injury	79	103	103	142	191
105B - Racially or Religiously Aggravated Assault Without Injury	2	3	1	8	4
10B - Possession Firearms Offences	1				
10C - Possession of Other Weapons	1		1		1
10D - Possession of Article with Blade or Point	5	1	1	5	11
126 - Vehicle Interference and Tampering			1		2
17A - Sexual Assault on a Male aged 13 and over			2	3	2
17B - Sexual Assault on a Male Child under 13			1		1
19C - Rape of a Female aged 16 and over	3	1	1		1

**Keep code and category
separate if possible**

'Tidy' structure

Should be able to pivot by any column, with 'Total' in values

Filename	Sheet name	Force	Hospital	Crime category	Crime sub-category	Crime code	Outcome code	Outcome description	Total	Year from	Year to
Merseyside.xlsx	2019-2020	Merseyside	NOT SPECIFIED	Burglary - Business and Community	NOT SPECIFIED	NOT SPECIFIED	15	POLICE - NAMED SUSPECT, VICTIM SUPPORTS BUT EVIDENTIAL DIFFICULTIES	1	2019	2020
Merseyside.xlsx	2019-2020	Merseyside	NOT SPECIFIED	Criminal Damage excluding Arson	NOT SPECIFIED	NOT SPECIFIED	15	POLICE - NAMED SUSPECT, VICTIM SUPPORTS BUT EVIDENTIAL DIFFICULTIES	7	2019	2020
Merseyside.xlsx	2019-2020	Merseyside	NOT SPECIFIED	Drug Possession and Use	NOT SPECIFIED	NOT SPECIFIED	15	POLICE - NAMED SUSPECT, VICTIM SUPPORTS BUT EVIDENTIAL DIFFICULTIES	13	2019	2020
Merseyside.xlsx	2019-2020	Merseyside	NOT SPECIFIED	Drug Trafficking	NOT SPECIFIED	NOT SPECIFIED	15	POLICE - NAMED SUSPECT, VICTIM SUPPORTS BUT EVIDENTIAL DIFFICULTIES	1	2019	2020
Merseyside.xlsx	2019-2020	Merseyside	NOT SPECIFIED	Miscellaneous Crimes Against Society	NOT SPECIFIED	NOT SPECIFIED	15	POLICE - NAMED SUSPECT, VICTIM SUPPORTS BUT EVIDENTIAL DIFFICULTIES	2	2019	2020
Merseyside.xlsx	2019-2020	Merseyside	NOT SPECIFIED	Other Sexual Offences	NOT SPECIFIED	NOT SPECIFIED	15	POLICE - NAMED SUSPECT, VICTIM SUPPORTS BUT EVIDENTIAL DIFFICULTIES	8	2019	2020

Prompt template for reshaping data

Break it down:

Objective

Task

Table structure

Extra details

Prompt pt1: Objective/context

Objective:

To reshape a CSV with data from an FOI response to a question on crimes in hospitals so that we have a single dataset in a common structure.

That structure has the following columns:

1. Filename
2. Sheet name/number
3. Force (extracted from the file name)
4. Crime category (where specified. If not specified, say 'NOT SPECIFIED'). For example 'Crime Tree LV4 Desc' is a higher level category but 'Stats Classification Desc' is a sub category to that.
5. Crime code (where specified. If not specified, say 'NOT SPECIFIED')
6. [ETC.]

Prompt pt2: Describe the task

Task

Look at the attached CSV and suggest Python code that will work in Colab notebook. It should:

1. Import the CSV
2. Reshape the data so it has the structure detailed above, separating any text that needs to go into different columns, such as category/outcome text and code, and year from/year to
3. Export as a CSV
4. Download the CSV

Provide the code in two blocks: the first block to upload the file, and the second block for everything after that point.

Comment the code thoroughly so that it is easy for a non-coder to understand. Avoid jargon.

Simple but longer code is preferable to shorter more complex code, if simple code is easier to understand.

Problem 6:
You need to clean
consistently

546 categories?

- Inconsistency of category description is a common challenge in FOI responses
- How do you get 546 categories down to... 12?

Pure vibe coding fail 2

Codex/Claude Code given 2 lookup files + 29 CSVs of clean responses OR list of 570 categories

Accuracy (matching the hybrid approach) ranged from **35%** (Codex with CSVs) to **53%** (Claude with list)

Triage by problem — again!

1. Mixed data (code and category; category and Act/Section)
 - a. Text to columns
 - b. XLOOKUP function
2. Acronyms (e.g. GBH = “Grievous bodily harm”)
3. Upper/lower case
 - a. LOWER or PROPER function (use IF+UPPER/REGEXMATCH to identify all-caps entries)
4. Extra/analogous characters: quotation marks, spaces, apostrophes, commas, “and”/“&”, currency symbols
 - a. Find and replace or SUBSTITUTE function; TRIM; REGEXREPLACE
5. Different category level (top level, sub-category, lower level)
 - a. XLOOKUP function; COUNTIF to identify keywords > manual assignment to category
6. Slight variations
 - a. Sort and manual edits; Open Refine: cluster and edit

Takeaways

- **Audit and design** a destination you are heading for (target structure)
- **Triage**: different extraction/cleaning problems
- **Combine** human and AI to reduce mistakes by both

Questions? Experiences?

Audit & prioritise

Audit responses to identify the level of detail in each response and identify edge cases. [Include a caveats column.](#)
Augment manual audit with **NotebookLM** audit.
Identify a priority order for data, e.g. totals by outcome, hospital, category or year where these are provided separately

Data structure

Design a data structure that can accommodate all responses
Structure should follow **'tidy' data principles**, i.e. one row per combination of features (force, category, hospital, outcome, year)
Structure should include source details, e.g. filename, sheet name, name of person entering data

Extract & reshape

PDFs: use **Tabula** or **vibe coding** (design a prompt template to generate code to attempt to extract data). Multi-sheet XLS files: use **Open Refine** to import and combine sheets
Design a **prompt template** for generating code to reshape CSV responses

Check extract, check reshape

Manual checks (e.g. compare entries, check page-ending rows)
Analysis-based checks (e.g. pivots, totals)
AI-based checks using a prompt template (e.g. compare files)

Combine

Use **OpenRefine** or: **Design a prompt template** for generating code to combine the resulting CSV files.

Bonus material

Meta prompting (more energy intensive):

Attached is data on crimes in hospitals.

Generate an expert prompt I can use in Copilot to extract data on the numbers of crimes in each category for each year.

You are working in the workbook "Leicestershire.xlsx". Build me a clean summary of crimes by category for each year using all the data in this file.

Data structure:

- "Key to Offence Codes & Outcome" (sheet): contains "Offence Code" and "Offence Description". Use this to map codes → readable categories.
- Yearly sheets named "19-20", "20-21", "21-22", "22-23", "23-24".
- In each yearly sheet, the "Offence Code / Total" block lists the offence code counts for that year. Ignore the Hospital and Outcome sections; I only want offence code counts.

Tasks:

1) For each yearly sheet:

- Identify the two-column block headed "Offence Code" and "Total" (some sheets have the header "Offence Code" with a separate "Total" column beneath it).
- Extract all rows where Offence Code is nonblank and Total is numeric.
- Create a normalized table with columns: [Year], [Offence Code], [Count].
 - Year values should be "2019/20", "2020/21", "2021/22", "2022/23", "2023/24" corresponding to sheets "19-20", "20-21", "21-22", "22-23", "23-24".
 - If the same offence code appears multiple times for a year, aggregate (sum) the counts.

2) Map offence codes to descriptions:

- Left-join the normalized table to the "Key to Offence Codes & Outcome" sheet using [Offence Code] → [Offence Code] to add [Offence Description].
- If a code has no description in the key, keep the code and label description as "Unknown".

3) Create a final summary matrix:

- Rows: Offence Description (use Offence Code as a secondary key to disambiguate—e.g., "Assault with injury (08N)").
- Columns: Year (2019/20 ... 2023/24).
- Values: Sum of Count.
- Include a "Grand Total" column and sort rows by Grand Total descending.

4) Output:

- Insert the summary as a new table named Crimes_By_Category_By_Year with formatted column headers.
- Also generate a clustered column chart (or heatmap-style conditional formatting) to visualize changes over time.
- Add a brief note above the table: "Source: Leicestershire.xlsx (Offence Code counts by year; mapped using Key to Offence Codes & Outcome)."
- Return any data-quality notes you detect (e.g., duplicate codes, blank totals, or codes without descriptions).

5) Deliverables:

- The normalized long table (Year, Offence Code, Offence Description, Count).
- The wide summary table as specified.
- The chart.

“Vibe coding”.

Original definition: letting an AI tool write code without any manual intervention

But often used to refer to any use of AI to generate code

Distinguished from “agentic engineering” where there is some quality control

“Good code” (Simon Willison)

- The code works. It does what it's meant to do, without bugs.
- We ***know the code works***. We've taken steps to confirm to ourselves and to others that the code is fit for purpose.
- It solves the right problem.
- It handles error cases gracefully and predictably
- It's simple and minimal
- It's protected by tests.
- It's documented at an appropriate level
- The design affords future changes
- All of the other relevant "ilities" - accessibility, testability, reliability, security, maintainability, observability, scalability, usability

“If an LLM wrote the code for you, and **you then reviewed it, tested it thoroughly and made sure you could explain how it works** to someone else that’s not vibe coding, it’s software development. The usage of an LLM to support that activity is immaterial.”

Simon Willison, *Not all AI-assisted programming is vibe coding (but vibe coding rocks)*



Ulvund Solstad (Economist) on vibe coding: <https://www.youtube.com/watch?v=3r4grJ5zwSQ&t=8465s>